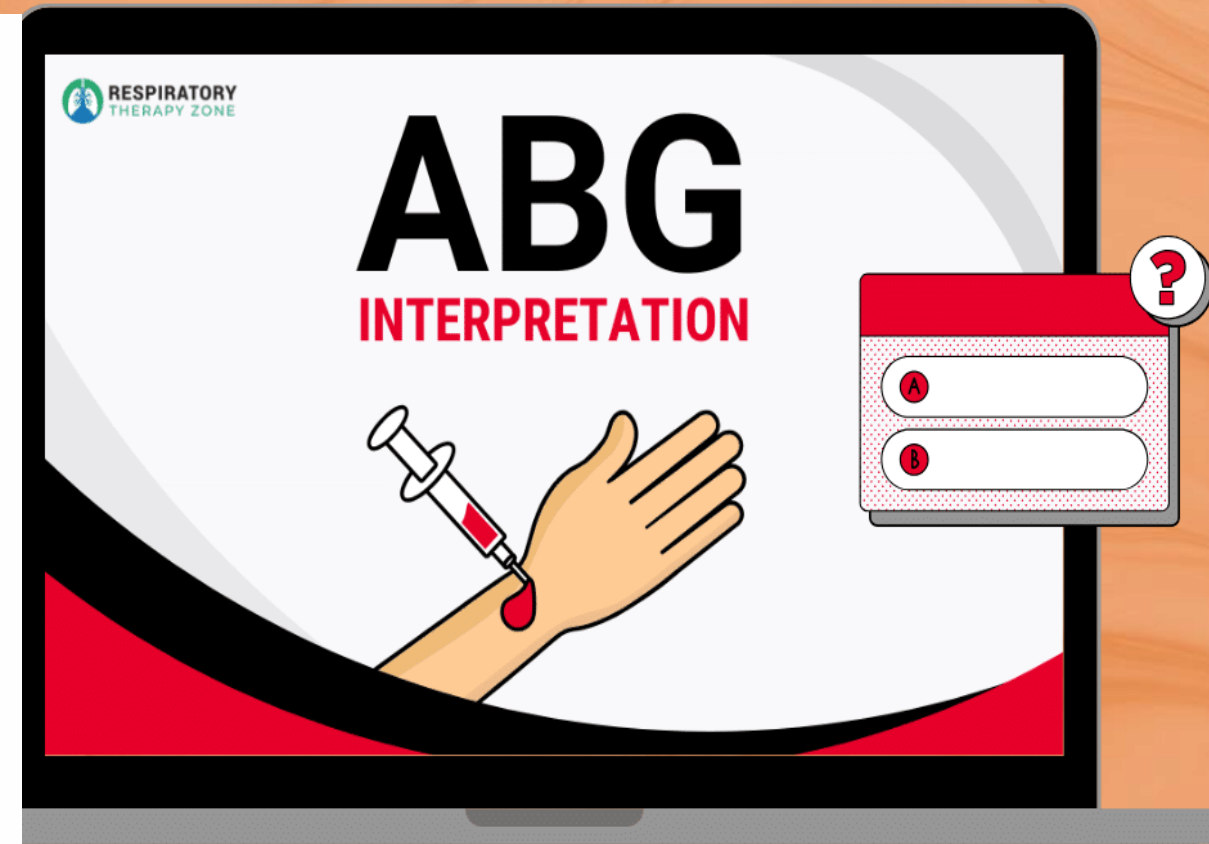
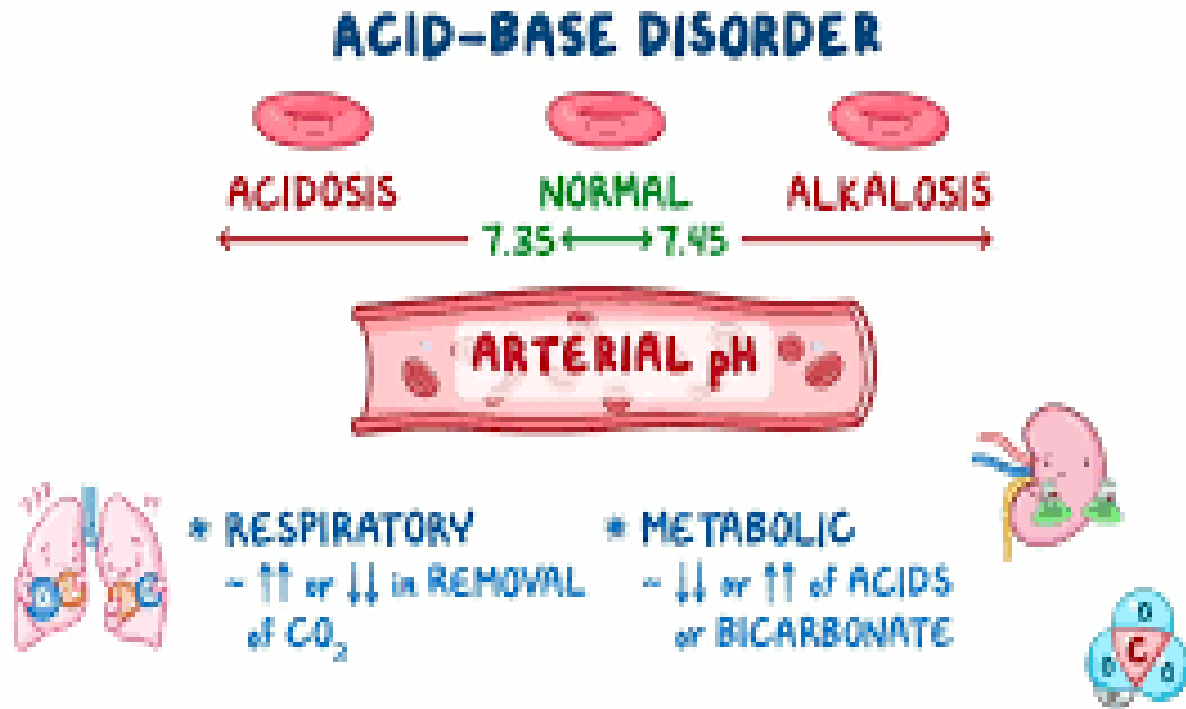


ABG interpretation & Acid base imbalance



Prepared by :

Assist professor Dr:Afaf Mohamed Mansour

Outline



1. components of buffering systems
2. ABG interpretation
3. **Alteration in acid base**
 1. **Respiratory acidosis**
 1. Physiological Respiratory acidosis
 2. Causes
 3. Clinical assessment findings for patients with Respiratory acidosis
 4. Management for Respiratory acidosis
 2. **Respiratory Alkalosis**
 1. Physiological Respiratory Alkalosis
 2. Causes
 3. Clinical assessment findings for patients with Respiratory Alkalosis
 4. Management for Respiratory Alkalosis

Outline

1. Alteration in acid base

1. Metabolic acidosis

1. Physiological Metabolic acidosis
2. Causes
3. Clinical assessment findings for patients with Metabolic acidosis
4. Management for Metabolic acidosis

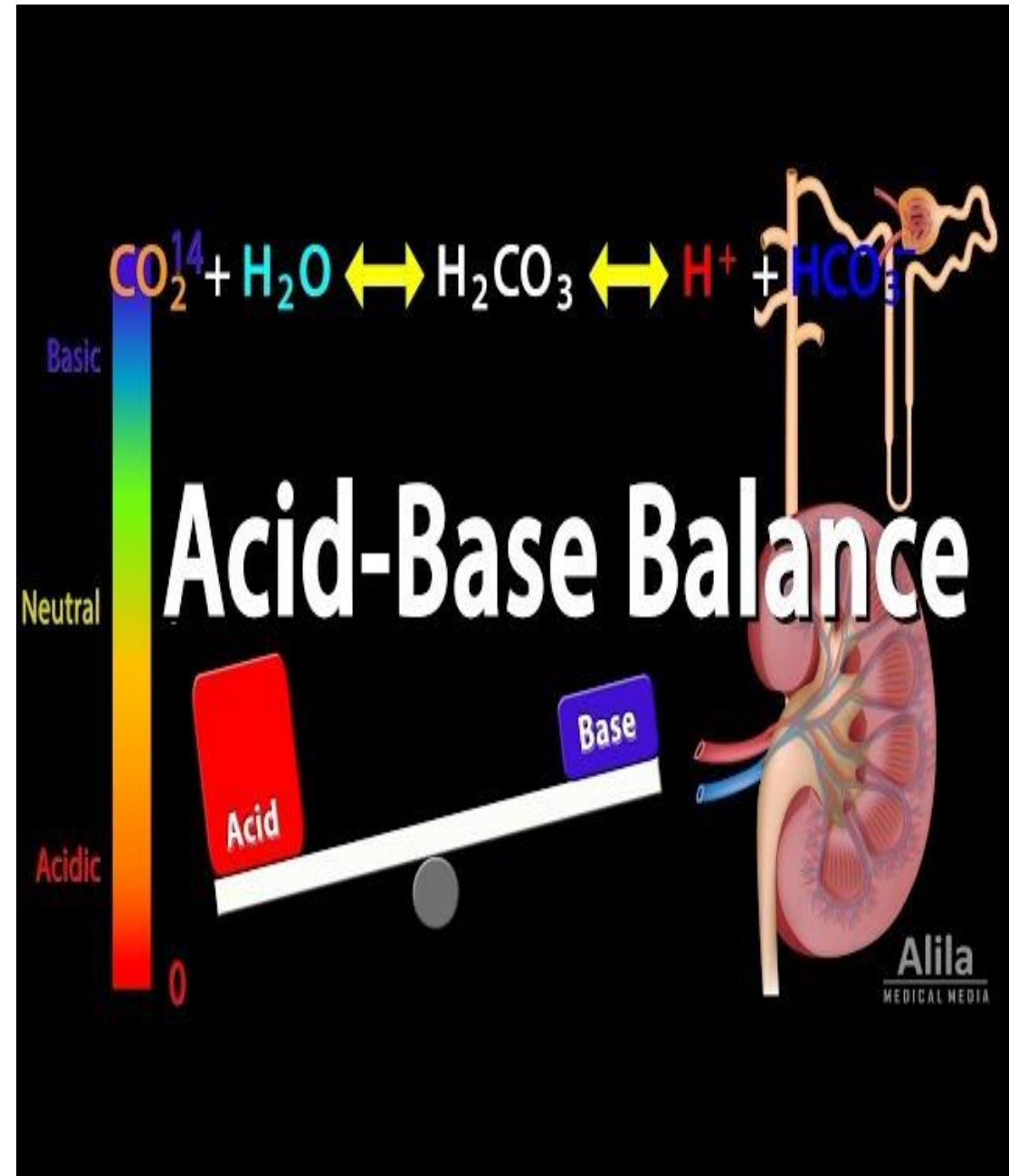
2. Metabolic Alkalosis

1. Physiological Metabolic Alkalosis
2. Causes
3. Clinical assessment findings for patients with Metabolic Alkalosis
4. Management for Metabolic Alkalosis

Buffer System

The body regulates acid-base balance through the buffer system. The buffer system can be described as a mechanism for the neutralization of acids.

Buffer system prevent major changes in the pH of body fluids by removing or releasing H^+ ; they can act quickly to prevent excessive changes in H^+ concentration.

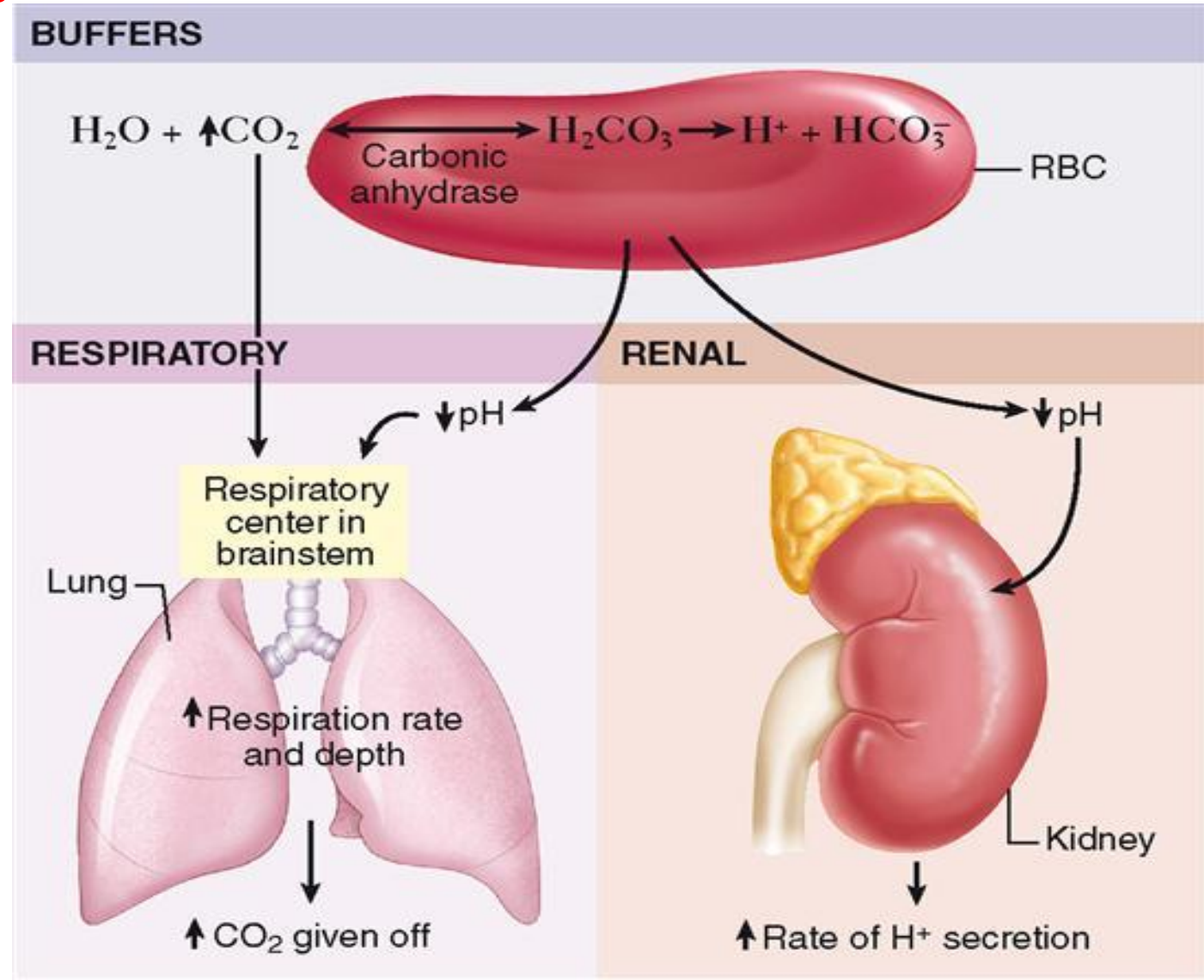


Three buffer systems exist for maintaining acid-base status:

*Blood

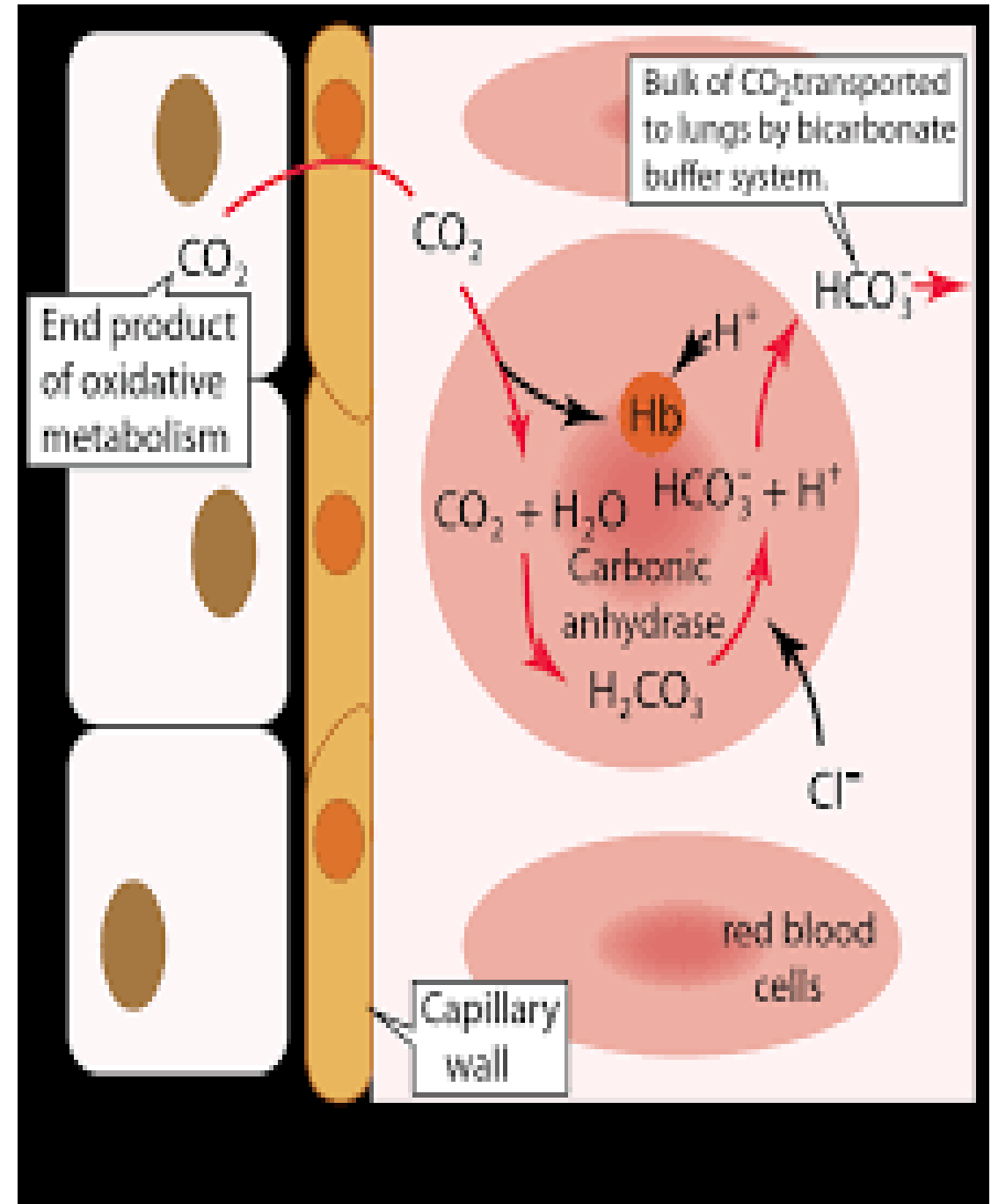
*Respiratory system

*Kidney



The Blood Buffer System

- Activated as the H^+ concentration changes.
As H^+ level increases, the pH falls, resulting in acidosis.
- Bicarbonate (HCO_3^-) then combines with H^+ to form carbonic acid (H_2CO_3).
- Carbonic acid then breaks down into carbon dioxide (which is **excreted through the lungs**) and water (H_2O).

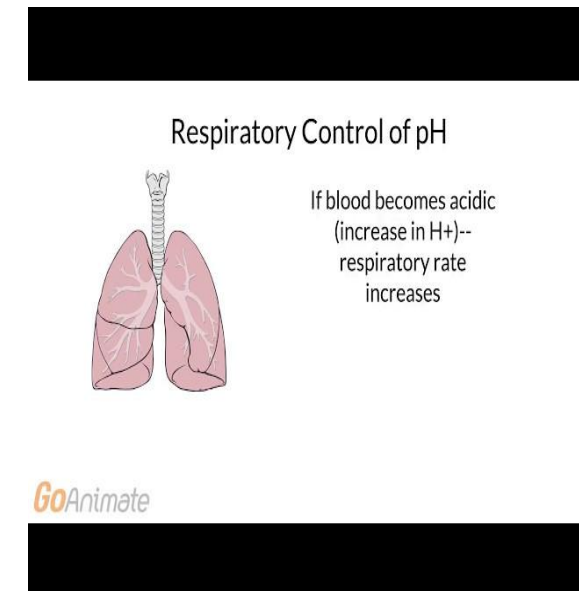
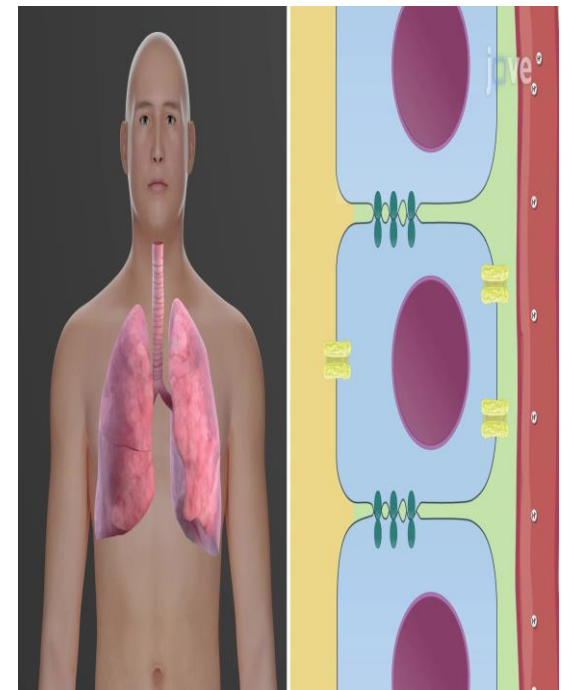


The Respiratory Buffer System

The lungs react to correct imbalance by altering the rate and depth of respiration.

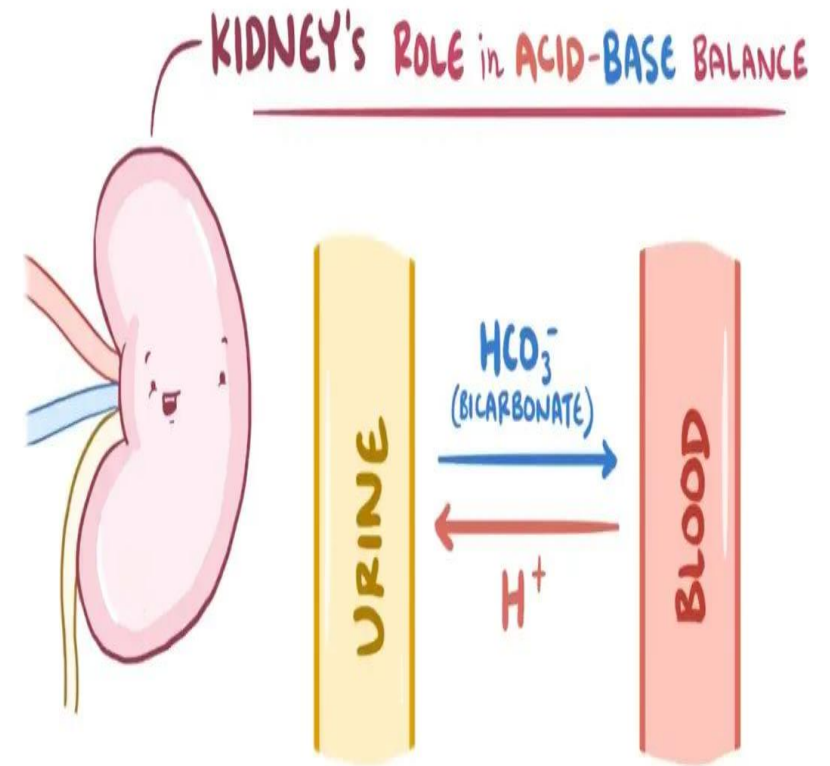
➤ **In metabolic acidosis:** the respiratory rate increase, resulting in a greater amount exhaled carbon dioxide which results in decrease acidic level by excreting excess carbon dioxide from the lungs (to reduce the acid load).

➤ **In metabolic alkalosis:** the respiratory rate decreases, causing CO₂ to be retained (to increase the acid load)



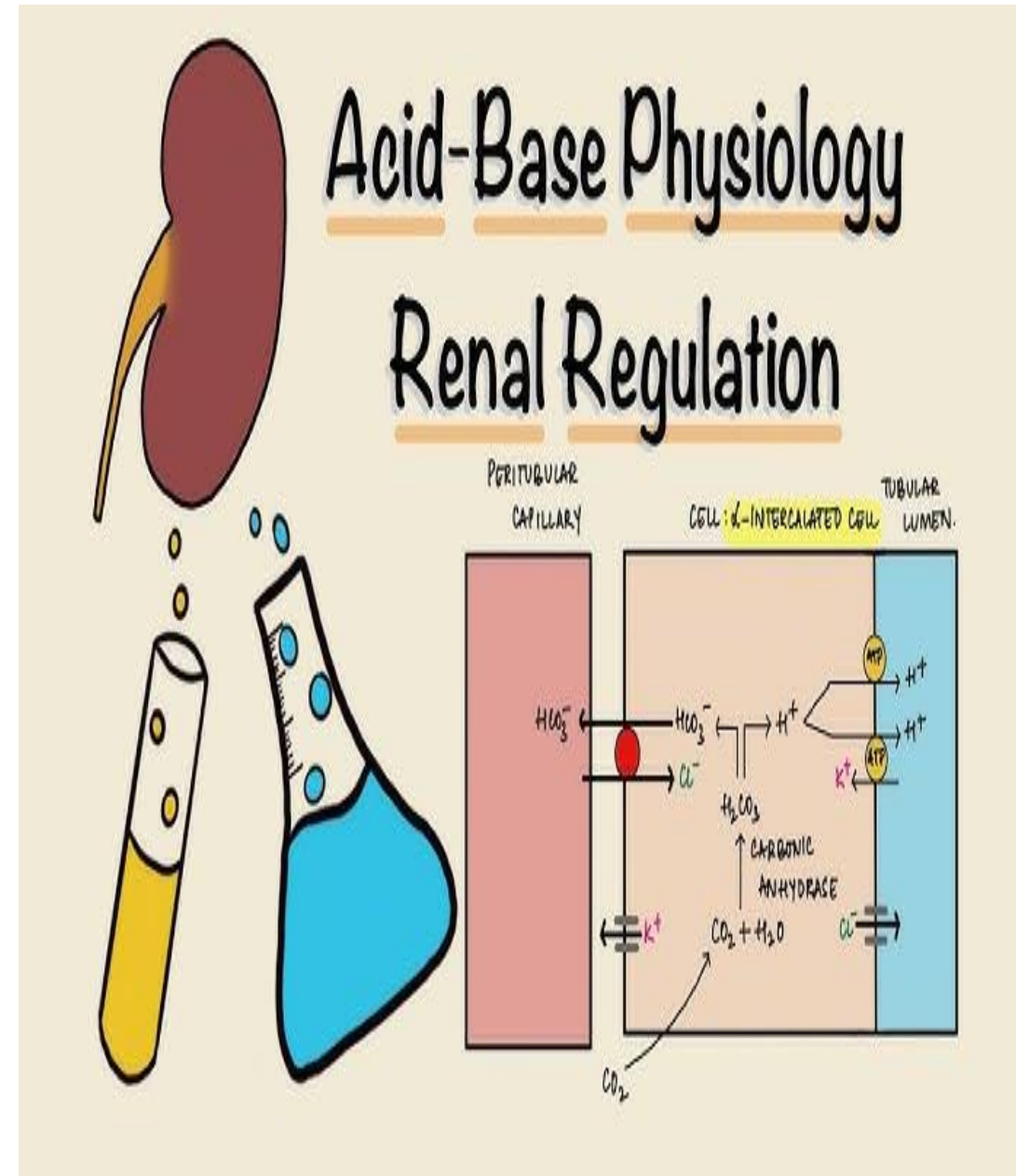
The Renal Buffer System

- The kidneys also play an important role in maintaining acid-base homeostasis by regulating the pH of the blood plasma.
- The kidneys have **two** very important roles in maintaining the acid-base balance:
 1. Reabsorb bicarbonate from urine
 2. Excrete hydrogen ions into urine.



The Renal Buffer System

- The kidneys are slower to compensate than the lungs; may take up to 2 days to regulate acid base balance.
- **In respiratory acidosis:** kidneys excrete hydrogen ions and conserve bicarbonate ions to help restore balance.
- **In respiratory alkalosis:** kidneys retain hydrogen ions and excrete bicarbonate ions



Steps in Arterial Blood Gas Interpretation

Arterial blood gas results should be interpreted systematically.

- ✚ The oxygenation is evaluated first.
- ✚ Second, the acid-base status is determined.
- ✚ Third, the primary imbalance is identified.
- ✚ Last, compensation is identified



Step 1: Evaluate Oxygenation

Oxygenation is analyzed by evaluation of the PaO₂ and the SaO₂. If the PaO₂ is less than the expected normal range, **hypoxemia** exists

paO₂ > 80	Normal
paO₂ <80	Hypoxia

Step 2: Evaluate Acid-Base Status

Evaluate the pH. A pH of less than 7.35 indicates the presence of acidemia; a pH of greater than 7.45 indicates presence of alkalemia.

pH <7.35	Acidotic
pH >7.45	Alkalotic

Evaluate the PaCO₂.

- ❖ A PaCO₂ of less than 35 mmHg indicates the presence of respiratory alkalosis or hyperventilation;
- ❖ a PaCO₂ of greater than 45 mm Hg indicates the presence of respiratory acidosis or hypoventilation.

	PH	Paco2	co3
Respiratory Acidosis	↓	↑	Normal
Respiratory Alkalosis	↑	↓	Normal

Evaluate the HCO₃

- ❖ An HCO₃ value of less than 22 Meq/L indicates the presence of metabolic acidosis;
- ❖ a HCO₃ of greater than 26 Meq/L indicates the presence of metabolic alkalosis.

	PH	Hco3	Paco2
Metabolic Acidosis	↓	↓	Normal
Metabolic Alkalosis	↑	↑	Normal

Step 3: Determine the primary acid-base Imbalance

The primary cause of the acid-base imbalance is determined by evaluation of the pH. If the pH is less than 7.35, the primary disorder is acidosis. If the pH is greater than 7.45, the primary disorder is alkalosis

Step 4: Determine Compensation

The body has three buffer systems that maintain a constant acid-base balance.

- Normal pH is fully compensated.
- Abnormal pH is partially compensated.

Example 1 : a 47-year-old kidney dialysis patient is admitted to the hospital. He has missed his last 2 dialysis treatments. His blood gas results are as follows:
What the interpretation of this result? pH 7.32, PaCO₂ 34, HCO₃ 18

Example 1: Steps in Arterial Blood Gas Interpretation

Step 1: Examine the pH level. It is low, therefore it is acidosis.

Step 2: Examine the PaCO₂ level. It is low. The pH and PaCO₂ are moving in the same direction; therefore this is a metabolic disorder. In this case, the lungs are now attempting to compensate by “blowing off” excessive CO₂, and therefore increasing the pH.

Step 3: Examine the HCO₃ level. It is low. This confirms that it is a metabolic issue.

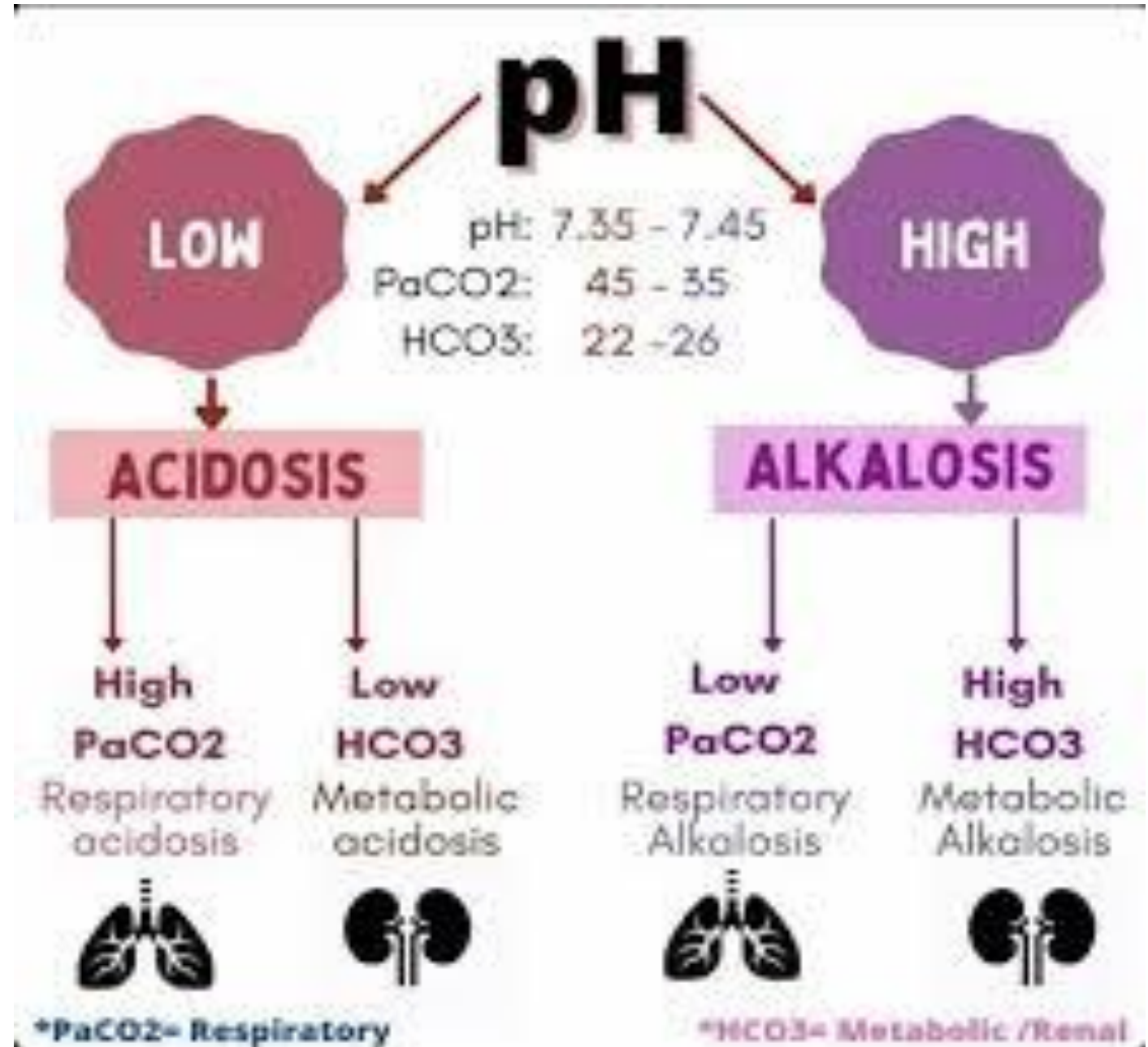
Interpretation = metabolic acidosis with partially compensated

ARTERIAL BLOOD GAS INTERPRETATION

1° DISORDER	pH	$P_a\text{CO}_2$	$[\text{HCO}_3^-]$	COMPENSATION
AG/non-AG Metabolic Acidosis	↓	↓ (2°)	↓ (1°)	$P_a\text{CO}_{2, \text{ expect}} = 1.5 [\text{HCO}_3^-] + 8 \pm 2$ If $P_a\text{CO}_{2, \text{ actual}} < P_a\text{CO}_{2, \text{ expect}}$ also 1° respiratory alkalosis If $P_a\text{CO}_{2, \text{ actual}} > P_a\text{CO}_{2, \text{ expect}}$ also 1° respiratory acidosis
AG Acidosis "Delta/Delta"	For AG metabolic acidosis, calculate $\Delta\text{AG} / \Delta[\text{HCO}_3^-] = (\text{AG} - 12) / (24 - [\text{HCO}_3^-])$ if < 0.8, non-AG acidosis ; if > 2, metabolic alkalosis			
Metabolic Alkalosis	↑	↑ (2°)	↑ (1°)	$P_a\text{CO}_2 = 0.7 \times [\text{HCO}_3^-] + 20 \pm 5$ If $P_a\text{CO}_{2, \text{ actual}} < P_a\text{CO}_{2, \text{ expect}}$ also 1° respiratory alkalosis If $P_a\text{CO}_{2, \text{ actual}} > P_a\text{CO}_{2, \text{ expect}}$ also 1° respiratory acidosis
Respiratory Acidosis	↓	↑ (1°)	↑ (2°)	For each ↑ 10 mmHg in $P_a\text{CO}_2$ Acute: ↑ $[\text{HCO}_3^-]$ 1 mmol/L and ↓ pH 0.08 Chronic: ↑ $[\text{HCO}_3^-]$ 4 mmol/L and ↓ pH 0.03
Respiratory Alkalosis	↑	↓ (1°)	↓ (2°)	For each ↓ 10 mmHg in $P_a\text{CO}_2$ Acute: ↓ $[\text{HCO}_3^-]$ 2 mmol/L and ↑ pH 0.08 Chronic: ↓ $[\text{HCO}_3^-]$ 5 mmol/L and ↑ pH 0.03
Primary disorder (1°), compensation (2°); arrows relative to "normal" baseline values: pH 7.35 - 7.45, $P_a\text{CO}_2$ 35 - 45 mmHg and $[\text{HCO}_3^-]$ 22 - 26 mEq/L				

Alteration in acid base

- Respiratory Acidosis
- Respiratory alkalosis
- Metabolic Acidosis
- Metabolic alkalosis



Respiratory Acidosis

- is a medical emergency in which decreased ventilation (hypoventilation) **increases** the concentration of **carbon dioxide** in the blood and **decreases the blood's pH** (a condition generally called acidosis).
- Alveolar hypoventilation thus leads to an increased PaCO_2 (a condition called hypercapnia). The increase in PaCO_2 in turn decreases the $\text{HCO}_3^-/\text{PaCO}_2$ ratio and decreases pH

Types of Respiratory Acidosis

Acute Respiratory Acidosis

The PaCO₂ is elevated above the upper limit of the reference range (over 45 mmHg) with an accompanying acidemia (pH <7.35).

- **Chronic Respiratory Acidosis,**

The PaCO₂ is elevated above the upper limit of the reference range, with a normal blood pH (7.35 to 7.45) or near-normal pH secondary to **renal compensation and an elevated serum bicarbonate (HCO₃ >30 mEq/l).**

Respiratory Acidosis  **Metabolic alkalosis**
Compensate

Causes of Respiratory Acidosis

1- Inhibition of the Medullary Respiratory Center

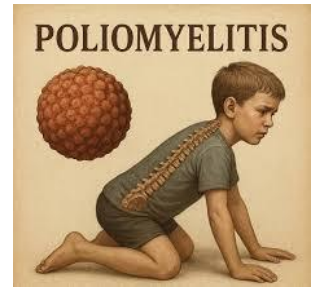
A. Acute causes:

- Drugs: anesthetics, sedatives
- Cardiac arrest

- Oxygen in chronic hypercapnia
- Central sleep apnea

B. Chronic causes:

- Extreme obesity (pickwickian syndrome)
- Metabolic alkalosis
- Central nervous system disease (rare)



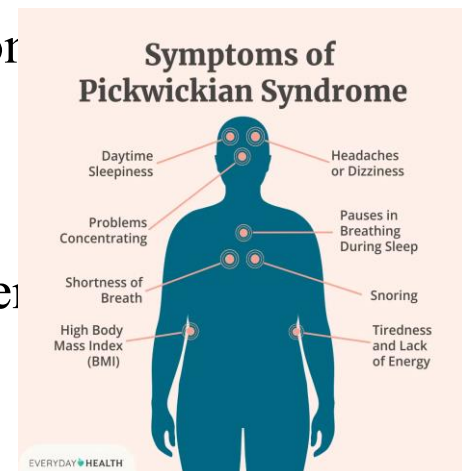
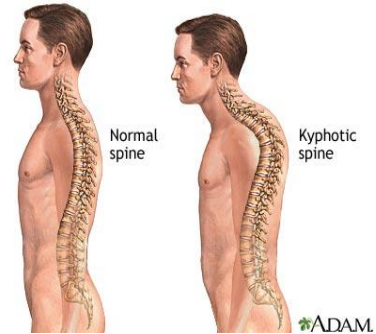
2- Disorders of the Respiratory Muscles and Chest Wall

A- Acute causes:

- Muscle weakness**: myasthenia gravis, periodic paralysis, -Guillain-Barré syndrome
- Severe hypokalemia -Hypophosphatemia

B. Chronic cause :

- Muscle weakness**: spinal cord injury, poliomyelitis, multiple sclerosis, and myxedema
- Kyphoscoliosis
- Extreme obesity



Causes of Respiratory Acidosis

3-Upper Airway obstruction

-Aspiration -Obstructive sleep apnea -laryngospasm Mechanical Ventilation

Permissive hypercapnia

Reversal of hyperventilation

4- Disorders affecting Gas Exchange Across the Pulmonary Capillary

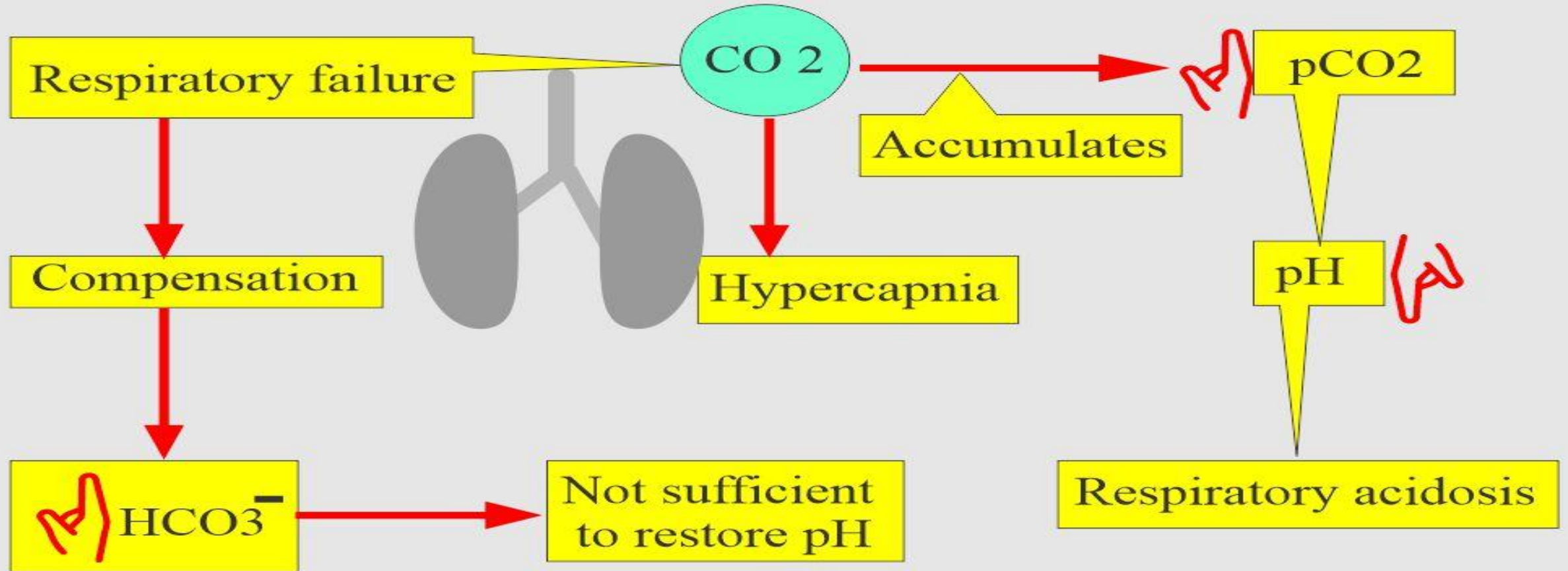
A. Acute

- Acute respiratory distress syndrome
- Acute cardiogenic pulmonary edema
- Severe asthma or pneumonia
- Pneumothorax /Pulmonary hemorrhage

B. Chronic:

COPD, bronchitis, emphysema and extreme obesity

Chronic respiratory acidosis



Total CO₂ rises to very high level in Chronic respiratory acidosis

Symptoms of Respiratory Acidosis

- **The initial symptoms include:**

1. Headache, blurred vision,
2. Restlessness, and anxiety, which can progress to tremors, asterixis, delirium, and somnolence (called CO₂ narcosis).
3. The cerebrospinal fluid pressure is often elevated
4. Papilledema may be seen.
5. Arrhythmias and peripheral vasodilatation can produce hypotension if the systemic pH falls below 7.10.
6. Rapid shallow respiration, muscle weakness.

Symptoms of Respiratory Acidosis

Chronic respiratory acidosis is also associated with:

1. Cor pulmonale and peripheral edema.
2. The cardiac output and glomerular filtration rate are usually normal in this disorder
3. Memory loss, sleep disturbances, and personality changes

Management of Respiratory Acidosis

Treating acute acidosis usually means addressing the underlying cause.

- **The criteria for admission to the intensive care unit (ICU) include**

1. Patient confusion,
2. Lethargy
3. Respiratory muscle fatigue, and a low pH (< 7.25). For example airway may need to be cleared. This must be done as soon as possible.

Artificial ventilation may also be needed.

Management of Respiratory Acidosis

• Infusion of sodium bicarbonate is rarely indicated. This measure may be considered after cardiopulmonary arrest with an extremely low pH ($< 7.0 - 7.10$). In most other situations, sodium bicarbonate has no role in the treatment of respiratory acidosis

Chronic Respiratory Acidosis

The goal is to improve airway function. Some strategies include:

- ☐ Antibiotics to treat infection
- ☐ Diuretics to reduce excess fluid affecting the heart and lungs
- ☐ Bronchodilators to expand the airways
- ☐ Corticosteroids to reduce inflammation
- ☐ Mechanical ventilation in severe cases

Respiratory Alkalosis

- Is a medical condition in which increased respiration **elevates the blood pH** beyond the normal range (7.35–7.45) with a concurrent **reduction in arterial levels of carbon dioxide.**

Types of Respiratory Alkalosis

There are two types of respiratory alkalosis: chronic and acute as a result of the 3-5 day delay in kidney compensation of the abnormality.

- ✓ **Acute respiratory alkalosis** occurs rapidly, have a high pH because the response of the kidneys is slow.
- ✓ **Chronic respiratory alkalosis** is a more long-standing condition; the kidneys have time to decrease the bicarbonate level

Causes of Respiratory Alkalosis

- Heart attack - Pain
- Anxiety
- Asthma - Chronic Obstructive Pulmonary Disease
- Infection - Pulmonary Embolism
- Pregnancy Stress - Pulmonary disorder
- Fever - Liver disease
- Heart disorder or other, including improper mechanical ventilation.
- Vocal cord paralysis (compensation for loss of vocal volume results in overbreathing/breathlessness).
- Salicylate poisoning (aspirin overdose)

Respiratory alkalosis

compensatory
mechanism

pH



HCO_3^-

pCO_2



No change
(may be normal)

Causes:

1. Hyperventilation
2. Pain
3. Anxiety

Kidney will excrete
increased amount of
 HCO_3^- to lower the
pH

Symptoms of Respiratory Alkalosis

- Tachypnea
- Dizziness
- Hypokalemia
- Lethargy & Confusion
- Dry mouth
- Tingling in the arms
- Heart palpitations
- Impaired neuromuscular function (e.g., paresthesias, tetany, tremor)
- Nausea & vomiting

Management of Respiratory Alkalosis

The treatment for respiratory alkalosis depends on the underlying cause.

1-Panic and anxiety-related causes

- Treating the condition is a matter of raising carbon dioxide levels in the blood.
- The following strategies and tips are useful for respiratory alkalosis caused by over breathing due to panic and anxiety.

Management of Respiratory Alkalosis

2. Breathe into a paper bag

- Fill the paper bag with carbon dioxide by exhaling into it.
- Breathe the exhaled air from the bag back into the lungs.
- Repeat this several times.

3- Reassurance

- The symptoms of respiratory alkalosis can be frightening. This often causes faster and deeper breathing, making things worse.
- Having a calm loved one provide reassurance could help get breathing under control.

Management of Respiratory Alkalosis

Restrict oxygen intake into the lungs

- To do this, try breathing while pursing the lips or breathing through one nostril.
- For the second approach to be useful, the mouth and the other nostril need to be covered.

Metabolic Acidosis

Bicarbonate deficit - blood concentrations of bicarb drop below 22mEq/L

- **Causes Metabolic Acidosis**

- Accumulation of acids (lactic acid or ketones)(**Diabetes ketoacidosis (DKA)**)
- Loss of bicarbonate through diarrhea or renal dysfunction
- **Kidney disease** is one of the most common causes of metabolic acidosis. Because kidneys are not able to remove enough acid from blood, causing a build-up in body.
 - CKD(chronic kidney disease
 - Acute kidney injury (AKI). Healthy kidneys remove natural waste products from blood, including acids.
- Hyperkalemia (high potassium levels) -Albuminuria
- -Smoking - Anemia (low red blood cell level

METABOLIC ACIDOSIS

↓ ↓ PH

↑ ACIDS

↓ HCO_3^- (Bicarbonate)

Production
or
Accumulation of Acids

Impaired
excretion of
Acids

- ① uncontrolled diabetes
Starvation → Beta hydroxy butyrate
Acetoacetate

DIABETIC KETOACIDOSIS

- ② Hypotension/Hypoxia → Impaired O_2 delivery → Anaerobic metabolism → Lactic Acidosis
- Carbon monoxide → Inhibition of oxidative phosphorylation → Lactic Acidosis
- Liver disease → Impaired Lactate clearance → Lactic Acidosis

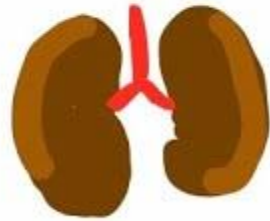
- ① Renal failure
- ② Distal Renal Tubular acidosis
- ③ Hypoadrenocorticism

Gastrointestinal Tract
Eg: Small bowel diarrhoea

HCO_3^- in diarrhoea > HCO_3^- in plasma

HYPERCHLOREMIC
METABOLIC
ACIDOSIS

Renal



Carbonic anhydrase inhibitor

Proximal
Renal Tubular
acidosis

**Inhibition of
oxidative phosphorylation**

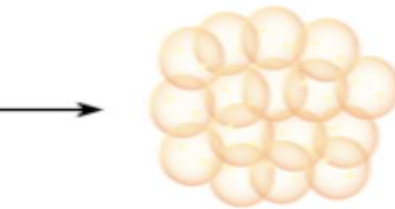


**Lactic acidosis
Hyperpyrexia**



**Increased renal
bicarbonate excretion**

2 Metabolic Acidosis



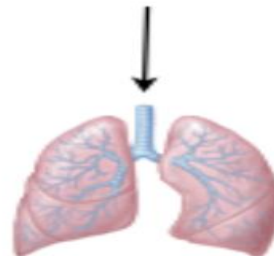
ketone bodies

**Increased fatty acid
metabolism**

wide anion gap

- Nausea, vomiting, diaphoresis
- Tinnitus, deafness, vertigo
- Agitation, delirium, hallucinations

Medullary stimulation



Hyperventilation

1 Respiratory Alkalosis

Predominates early

Symptoms of Metabolic Acidosis

- Headache, lethargy
- Nausea, vomiting, diarrhea
- Coma -Death
- **Compensation for Metabolic Acidosis**
 - Increased ventilation
 - Renal excretion of hydrogen ions if possible
 - K^+ exchanges with excess H^+ in ECF ,(H^+ into cells, K^+ out of cells)

Diagnosis of Metabolic Acidosis

- The diagnosis of metabolic acidosis is established from;
 1. Health history,
 2. Clinical symptoms,
 3. Laboratory findings: arterial $\text{PH} < 7.35$ and bicarbonate concentration $< 24 \text{ mEq/L}$

Management of Metabolic Acidosis

- IV lactate solution
- In order to effectively treat metabolic acidosis, the causative process must be identified and treated.
- The role of adjunctive therapy with sodium bicarbonate (NaHCO_3) is not universally agreed upon.
- The metabolic acidosis seen with lactic acidosis and ketoacidosis generally resolves with therapy targeted at the underlying cause and NaHCO_3 may be unnecessary regardless of the pH.
- The metabolic acidosis of renal failure, renal tubular acidosis, or intoxication with ethylene glycol, methanol, or salicylates is much more likely to require NaHCO_3 therapy
- Another option for patients with severe acidemia is **tromethamine (THAM)**.



Metabolic alkalosis:

Bicarbonate excess - concentration in blood is greater than 26 mEq/L

Causes of metabolic alkalosis:

- Excess vomiting = loss of stomach acid
- Certain diuretics
- Heavy ingestion of antacids
- Excessive use of alkaline drugs
- Endocrine disorders
- Severe dehydration
- **Compensation for Metabolic Alkalosis**
 - Alkalosis most commonly occurs with renal dysfunction, so can't count on kidneys
 - Respiratory compensation difficult – hypoventilation limited by hypoxia

METABOLIC ALKALOSIS



$\text{pH} > 7.45$

$\text{HCO}_3^- > 27 \text{ mEq}$
NORMAL
 $22 - 27 \text{ mEq}$

COMMON CAUSES



**PROLONGED
VOMITING**



HYPOVOLEMIA



**DIURETIC
USE**

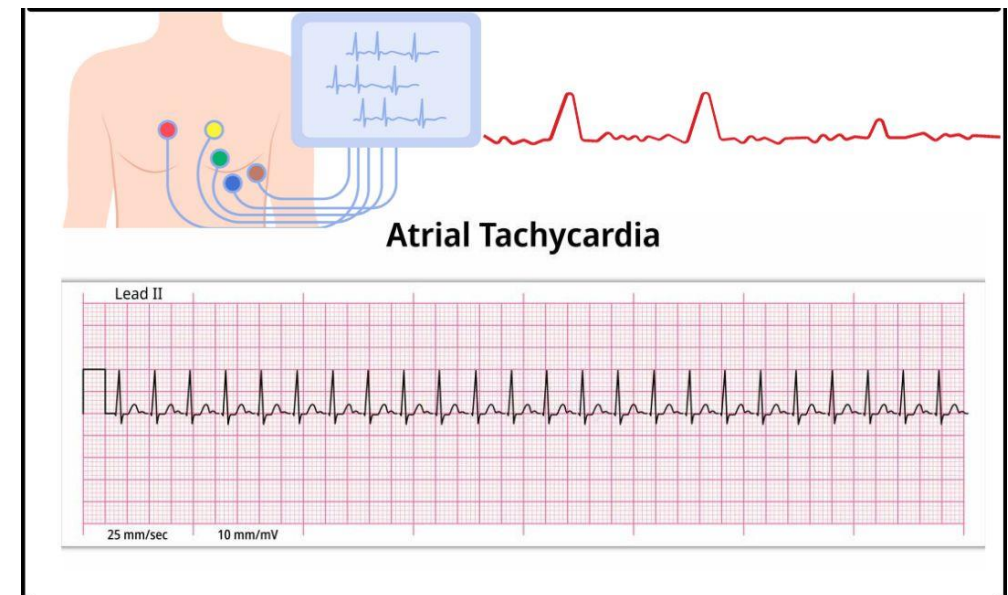
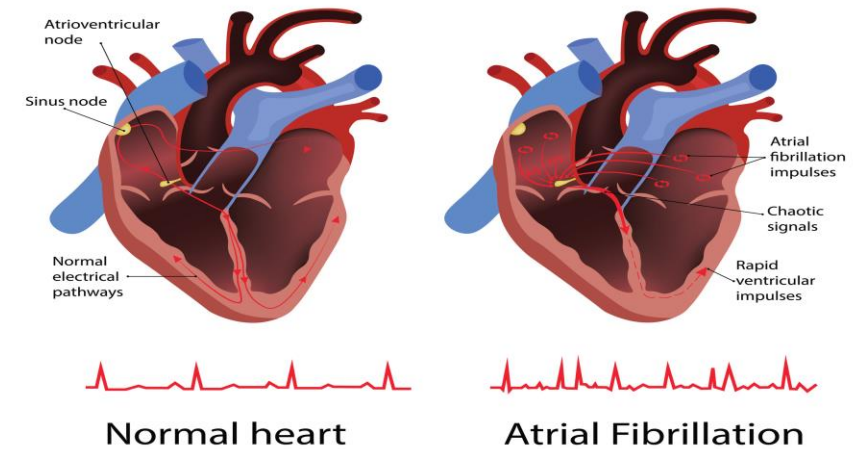


HYPOKALEMIA

Symptoms of Metabolic Alkalosis

- Respiration slow and shallow
- Hyperactive reflexes ; tetany Often related to depletion of electrolytes
- Atrial tachycardia
- Dysrhythmias

Heart arrhythmia



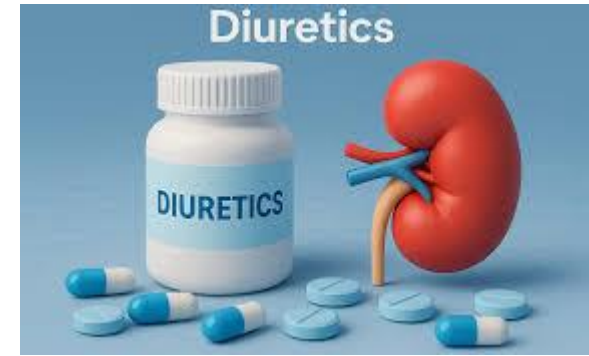
Management of Metabolic Alkalosis

- In order to effectively treat metabolic alkalosis, the *causative process must be identified and treated.*
- Always look for administration of compounds such as **citrate** in blood products and **acetate** in parenteral nutrition that can raise the HCO_3^- level.



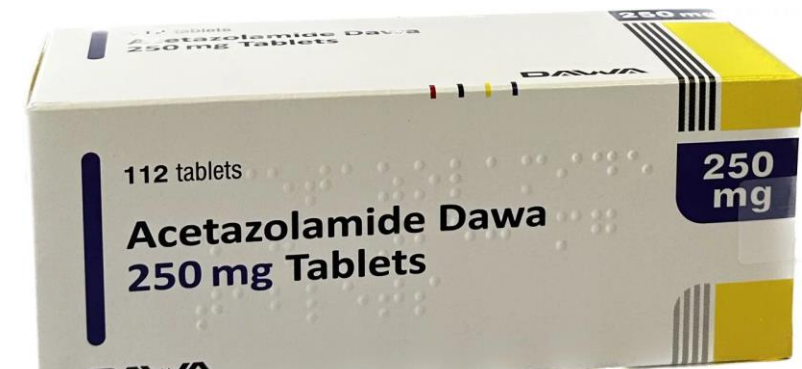
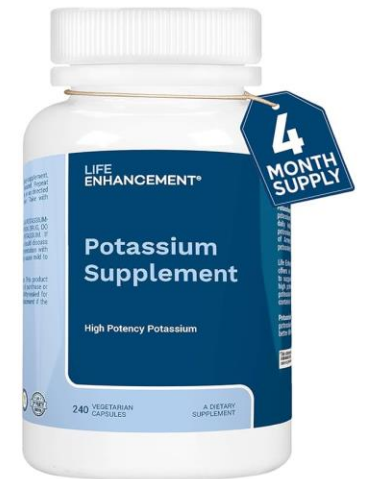
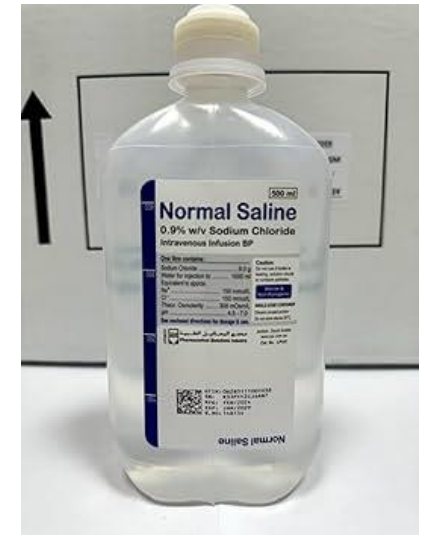
Management metabolic alkalosis

- If the etiology of the metabolic alkalosis is still unclear, measurement of the urinary chloride may be useful.
- ✓ Some processes leading to metabolic alkalosis (vomiting, nasogastric suction losses, and factitious diarrhea) will have low urinary Cl^- levels (less than 25 mEq/L or mmol/L)
- ✓ While others (diuretics, hypokalemia, and mineralocorticoid excess) will have higher urinary Cl^- levels (greater than 40 mEq/L or mmol/L).
- In general, contributing factors such as **diuretics, nasogastric suction, and corticosteroids** should be discontinued if possible



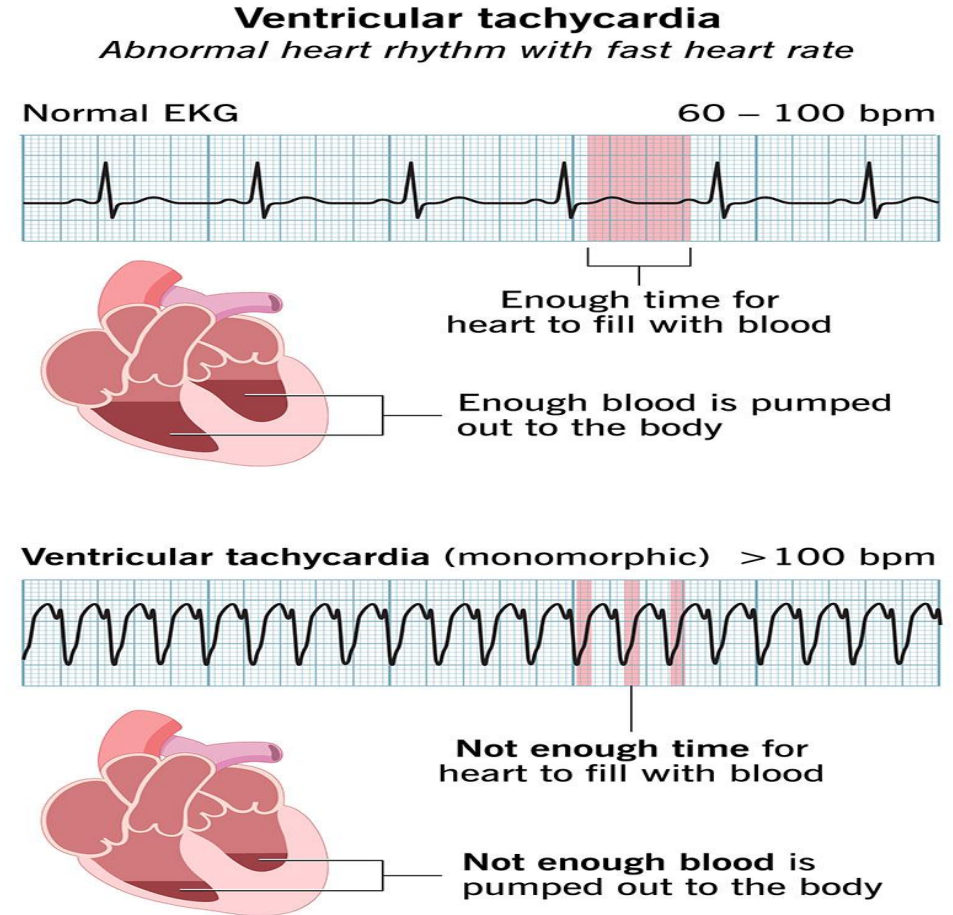
Management metabolic alkalosis

- Any fluid deficits should be treated with IV normal saline.
- Potassium supplementation should always be given if it is also deficient.
- In patients with mild or moderate alkalosis who require ongoing diuresis but have rising HCO_3^- levels, the carbonic anhydrase inhibitor acetazolamide can be used to reduce the HCO_3^- concentration. Acetazolamide is typically dosed at 250 mg every 6 to 12 hours as needed to maintain the pH in a clinically acceptable range.



Management metabolic alkalosis

- If alkalosis is profound and potentially life-threatening (due to seizures or ventricular tachyarrhythmias) consideration can be given to **hemodialysis**.





Q :differentiate between ?

items	• Respiratory Acidosis	• Respiratory alkalosis
PH		
paco2		
Causes		
Clinical manifestation		
Management		

items	• Metabolic Acidosis	• Metabolic alkalosis
PH		
HCO3		
Causes		
Clinical manifestation		
Management		

